

Homework #2

Answer the following questions. Submit your homework by e-mail to dataorg2017@gmail.com until December, 12th at 23:59.

Questions

1. Apply the bulk loading algorithm for the records having key values **20, 25, 10, 1, 16, 8, 3, 21, 22, 15, 9, 40, 45, 60, 13** to form a B+ tree of degree 2 (both for the index nodes and leaf nodes).
2. Insert **2** and **4** to the B+ tree that you have formed in the 1st question. And draw the B+ tree after each insertion.
3. Delete **1, 3** and **8** from the B+ tree that you have formed in the 1st question. And draw the B+ tree after each deletion.
4. Insert the records with the keys **2, 15, 23, 6, 35, 13, 8, 26, 1**, and **11** into a hash table using **extendible hashing**. Use **$h(\text{key}) = \text{key}$** as the hash function. Assume that global depth = 1, and the hash table is empty at the beginning. Each data page may hold up to 3 records. Show each step of the insertion.
5. Construct a grid file using the following data with **age as the first dimension** and **salary as the second dimension** etc. Let the age dimension range from 0 to 60 and the salary dimension range from 0 to 8000. Assume that each bucket has a capacity of two records. Show each step of the insertion.

Name	Age	Salary
Derya	28	3500
Aslı	35	5000
Jale	18	2700
Deniz	27	3500
Mehmet	45	5500
Engin	55	5500
Sami	19	2600
Rüya	50	5500